

Artificial Intelligence

Complete Study Notes — For RAG Testing

Prepared for: RAG (Retrieval-Augmented Generation) evaluation

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1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is the branch of computer science that focuses on building systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, perception, and language understanding. The term was coined by John McCarthy in 1956 at the Dartmouth Conference, widely considered the founding event of AI as an academic field.

Modern AI systems are broadly evaluated on the Turing Test, proposed by Alan Turing in 1950, which asks whether a machine's responses in conversation can be distinguished from a human's.

2. History of AI

Era	Period	Key Development
Foundations	1950–1956	Turing Test proposed; term 'AI' coined at Dartmouth.
Early Enthusiasm	1956–1974	First AI programs (Logic Theorist, ELIZA) built.
First AI Winter	1974–1980	Funding cuts after unmet expectations.
Expert Systems	1980–1987	Rule-based expert systems adopted commercially.
Second AI Winter	1987–1993	Expert systems' limitations become apparent.
Machine Learning Rise	1993–2011	Statistical ML, SVMs, and data-driven methods grow.
Deep Learning Era	2012–2020	AlexNet (2012) triggers the deep learning boom.
Generative AI Era	2020–present	Large language models (GPT, Claude, Gemini) emerge.

3. Types of AI

By capability

- Narrow AI (Weak AI): Designed for a single task, e.g. spam filters, voice assistants. All AI systems in wide use today are narrow AI.
- General AI (AGI): A hypothetical system with human-level reasoning across any task. Does not exist yet.
- Super AI (ASI): A hypothetical system that surpasses human intelligence in all domains.

By functionality

- Reactive Machines: React to current input with no memory of the past (e.g. Deep Blue chess engine).
- Limited Memory: Use past data to inform decisions (e.g. self-driving cars).
- Theory of Mind: Would understand emotions and beliefs of others — still research stage.
- Self-Aware AI: A hypothetical AI with consciousness — purely theoretical today.

4. Machine Learning Basics

Machine Learning (ML) is a subset of AI in which systems learn patterns from data rather than being explicitly programmed with rules. A typical ML workflow includes data collection, preprocessing, model training, evaluation, and deployment.

Common algorithms

- Linear Regression — predicts a continuous numeric value.
- Logistic Regression — used for binary classification problems.
- Decision Trees and Random Forests — rule-based, interpretable models.
- Support Vector Machines (SVMs) — find the optimal boundary between classes.
- K-Nearest Neighbours (KNN) — classifies based on the closest labeled examples.

5. Types of Machine Learning

Type	Description	Example
Supervised Learning	Learns from labeled input-output pairs.	Email spam detection
Unsupervised Learning	Finds patterns in unlabeled data.	Customer segmentation
Semi-Supervised Learning	Uses a mix of labeled and unlabeled data.	Medical image tagging
Reinforcement Learning	Learns via trial and error using rewards.	Game-playing agents (AlphaGo)

6. Deep Learning & Neural Networks

Deep Learning is a subfield of ML that uses artificial neural networks with many layers ('deep' networks) to automatically learn hierarchical feature representations from raw data such as images, audio, or text.

Key architectures

- CNN (Convolutional Neural Network) — primarily used for image and video processing.
- RNN (Recurrent Neural Network) — designed for sequential data such as time series or text.
- LSTM (Long Short-Term Memory) — an RNN variant that handles long-range dependencies better.
- Transformer — introduced in the 2017 paper 'Attention Is All You Need'; the architecture behind almost all modern large language models.

7. Natural Language Processing (NLP)

NLP is the field of AI concerned with enabling computers to understand, interpret, and generate human language. Common NLP tasks include sentiment analysis, machine translation, named entity recognition, summarization, and question answering.

Key concepts

- Tokenization — splitting text into smaller units (words or sub-words) for processing.
- Embeddings — dense numeric vector representations of words or sentences that capture meaning.
- Large Language Models (LLMs) — transformer-based models trained on massive text corpora to generate and understand language, e.g. GPT, Claude, Gemini, LLaMA.

8. Computer Vision

Computer Vision enables machines to interpret and process visual information from the world, such as images and video. Common tasks include image classification, object detection, semantic segmentation, and facial recognition.

Popular architectures include CNNs, ResNet, YOLO (You Only Look Once) for real-time object detection, and Vision Transformers (ViT), which apply the transformer architecture originally built for text to image patches.

9. Robotics and AI

Robotics combines AI with mechanical engineering to build physical agents that can sense and act in the real world. AI contributes perception (via computer vision and sensor fusion), planning (deciding a sequence of actions), and control (executing precise movement).

Notable examples include autonomous vehicles, industrial robotic arms, warehouse robots (e.g. Amazon's Kiva/Proteus systems), and humanoid robots such as Boston Dynamics' Atlas.

10. Generative AI & Large Language Models

Generative AI refers to systems that can create new content — text, images, audio, or code — rather than only classifying or predicting from existing data. Large Language Models (LLMs) are the most prominent example, trained via next-token prediction on vast text datasets and refined using techniques such as Reinforcement Learning from Human Feedback (RLHF).

Retrieval-Augmented Generation (RAG)

RAG is a technique that combines an LLM with an external retrieval system (often a vector database). Instead of relying only on knowledge learned during training, the model retrieves relevant document chunks at query time and uses them as additional context, which reduces hallucination and allows answers grounded in up-to-date or domain-specific information.

11. AI in Healthcare

- Medical imaging — AI models assist radiologists in detecting tumors, fractures, and abnormalities in X-rays, MRIs, and CT scans.
- Drug discovery — AI accelerates the search for candidate molecules, as demonstrated by DeepMind's AlphaFold for protein structure prediction.
- Diagnosis support — AI systems flag potential conditions from symptoms and patient history, though final decisions remain with clinicians.

12. AI in Finance

- Fraud detection — models flag unusual transaction patterns in real time.
- Algorithmic trading — AI systems execute trades based on statistical models and market signals at speeds beyond human capability.
- Credit scoring — ML models assess loan applicant risk using historical repayment data.

13. AI in Education

- Personalized learning — adaptive platforms adjust difficulty based on a student's performance.
- Automated grading — AI assists with grading essays and objective assessments.
- AI tutors and chatbots — provide 24/7 assistance for student questions and practice.

14. Ethics and Bias in AI

AI systems can inherit and amplify biases present in their training data, leading to unfair outcomes across race, gender, or socioeconomic lines. Well-known cases include biased facial recognition accuracy across skin tones and biased hiring algorithms.

Core ethical principles

- Fairness — ensuring AI decisions do not systematically disadvantage particular groups.
- Transparency — making model decisions explainable, especially in high-stakes domains.
- Accountability — establishing clear responsibility when an AI system causes harm.
- Privacy — protecting personal data used to train and operate AI systems.

15. AI Safety and Alignment

AI alignment research focuses on ensuring advanced AI systems act in accordance with human values and intentions, even as capabilities increase. Key concerns include reward hacking (an AI optimizing a proxy metric in unintended ways), robustness to adversarial inputs, and maintaining meaningful human oversight over powerful systems.

16. Future of AI

- Multimodal AI — models that jointly understand text, images, audio, and video are becoming standard rather than the exception.
- AI agents — systems that can autonomously plan and execute multi-step tasks using tools, rather than answering single queries.
- Regulation — governments worldwide (e.g. the EU AI Act) are introducing frameworks to govern AI development and deployment.

17. Career in AI

Common roles

Role	Focus
Machine Learning Engineer	Building and deploying ML models in production.
Data Scientist	Analyzing data and building predictive models.
NLP Engineer	Building language-focused AI systems.
AI Research Scientist	Advancing the theoretical and algorithmic frontier of AI.
MLOps Engineer	Managing infrastructure for training and serving ML models at scale.

18. Glossary

Term	Definition
Algorithm	A step-by-step procedure used to solve a problem or perform a computation.
Model	The output of a training process that can make predictions on new data.
Training	The process of adjusting a model's parameters using data.
Inference	Using a trained model to make predictions on new, unseen data.
Overfitting	When a model performs well on training data but poorly on new data.
Hallucination	When an AI model generates confident but factually incorrect output.

19. Frequently Asked Questions

Q: Is AI the same as Machine Learning?

A: No. AI is the broader field of building intelligent systems; Machine Learning is one approach within AI where systems learn from data.

Q: What makes RAG useful for testing AI systems?

A: RAG separates what a model 'knows' from what it retrieves at query time, so a well-designed test document with clear facts (like this one) lets you check whether the system retrieves the right chunk and grounds its answer in it, rather than guessing from general training knowledge.

Q: Does true Artificial General Intelligence (AGI) exist today?

A: No. As of this document, all deployed AI systems are Narrow AI; AGI remains a research goal, not a deployed reality.